

Problem 18.8

A benefactor leaves an inheritance to four charities, A , B , C , and D . The total inheritance is a series of level payments at the end of each year forever. During the first n years A , B , and C share each payment equally. All payments after n years revert to D . If the present values of the shares of A , B , C , and D are all equal, find $(1 + i)^n$.

Solution

Let each total annual payment be X . During the first n years, charities A , B , and C each receive

$$\frac{X}{3}$$

at the end of each year. Therefore, the present value of each of A , B , and C is

$$\frac{X}{3} a_{\overline{n}|i}.$$

Charity D receives all payments after the first n years. This is a deferred perpetuity-immediate, so the present value of D 's share is

$$Xv^n a_{\infty|i}.$$

Since all present values are equal,

$$\frac{X}{3} a_{\overline{n}|i} = Xv^n a_{\infty|i}.$$

Cancel X and use

$$a_{\overline{n}|i} = \frac{1 - v^n}{i} \quad \text{and} \quad a_{\infty|i} = \frac{1}{i}.$$

Then

$$\frac{1}{3} \left(\frac{1 - v^n}{i} \right) = v^n \left(\frac{1}{i} \right).$$

Multiplying by i ,

$$\frac{1 - v^n}{3} = v^n.$$

Thus,

$$1 - v^n = 3v^n.$$

So

$$1 = 4v^n,$$

and hence

$$v^n = \frac{1}{4}.$$

Since $v^n = (1 + i)^{-n}$,

$$(1 + i)^n = 4.$$

Therefore,

$$\boxed{(1 + i)^n = 4}.$$